



Chemotherapy with or without avelumab followed by avelumab maintenance versus chemotherapy alone in patients with previously untreated epithelial ovarian cancer (JAVELIN Ovarian 100): an open-label, randomised, phase 3 trial

Bradley J Monk*, Nicoletta Colombo, Amit M Oza, Keiichi Fujiwara, Michael J Birrer, Leslie Randall, Elena V Poddubskaya, Giovanni Scambia, Yaroslav V Shparyk, Myong Cheol Lim, Snehal Kumar M Bhoola, Joohyuk Sohn, Kan Yonemori, Ross A Stewart, Xiaoxi Zhang, Julia Perkins Smith, Carlos Linn, Jonathan A Ledermann*

Summary

Background Although most patients with epithelial ovarian cancer respond to frontline platinum-based chemotherapy, around 70% will relapse within 3 years. The phase 3 JAVELIN Ovarian 100 trial compared avelumab (anti-PD-L1 monoclonal antibody) in combination with chemotherapy followed by avelumab maintenance, or chemotherapy followed by avelumab maintenance, versus chemotherapy alone in patients with treatment-naïve epithelial ovarian cancer.

Methods JAVELIN Ovarian 100 was a global, open-label, three-arm, parallel, randomised, phase 3 trial run at 159 hospitals and cancer treatment centres in 25 countries. Eligible women were aged 18 years and older with stage III–IV epithelial ovarian, fallopian tube, or peritoneal cancer (following debulking surgery, or candidates for neoadjuvant chemotherapy), and had an Eastern Cooperative Oncology Group performance status of 0 or 1. Patients were randomly assigned (1:1:1) via interactive response technology to receive chemotherapy (six cycles; carboplatin dosed at an area under the serum-concentration-time curve of 5 or 6 intravenously every 3 weeks plus paclitaxel 175 mg/m² every 3 weeks or 80 mg/m² once a week [investigators' choice]) followed by avelumab maintenance (10 mg/kg intravenously every 2 weeks; avelumab maintenance group); chemotherapy plus avelumab (10 mg/kg intravenously every 3 weeks) followed by avelumab maintenance (avelumab combination group); or chemotherapy followed by observation (control group). Randomisation was in permuted blocks of size six and stratified by paclitaxel regimen and resection status. Patients and investigators were masked to assignment to the two chemotherapy groups without avelumab at the time of randomisation until completion of the chemotherapy phase. The primary endpoint was progression-free survival assessed by blinded independent central review in all randomly assigned patients (analysed by intention to treat). Safety was analysed in all patients who received at least one dose of study treatment. This trial is registered with ClinicalTrials.gov, NCT02718417. The trial was fully enrolled and terminated at interim analysis due to futility, and efficacy is no longer being assessed.

Findings Between May 19, 2016 and Jan 23, 2018, 998 patients were randomly assigned (avelumab maintenance n=332, avelumab combination n=331, and control n=335). At the planned interim analysis (data cutoff Sept 7, 2018), prespecified futility boundaries were crossed for the progression-free survival analysis, and the trial was stopped as recommended by the independent data monitoring committee and endorsed by the protocol steering committee. Median follow-up for progression-free survival for all patients was 10·8 months (IQR 7·1–14·9); 11·1 months (7·0–15·3) for the avelumab maintenance group, 11·0 months (7·4–14·5) for the avelumab combination group, and 10·2 months (6·7–14·0) for the control group. Median progression-free survival was 16·8 months (95% CI 13·5–not estimable [NE]) with avelumab maintenance, 18·1 months (14·8–NE) with avelumab combination treatment, and NE (18·2 months–NE) with control treatment. The stratified hazard ratio for progression-free survival was 1·43 (95% CI 1·05–1·95; one-sided p=0·99) with the avelumab maintenance regimen and 1·14 (0·83–1·56; one-sided p=0·79) with the avelumab combination regimen, versus control treatment. The most common grade 3–4 adverse events were anaemia (69 [21%] patients in the avelumab maintenance group, 63 [19%] in the avelumab combination group, and 53 [16%] in the control group), neutropenia (91 [28%], 99 [30%], and 88 [26%]), and neutrophil count decrease (49 [15%], 45 [14%], and 59 [18%]). Serious adverse events of any grade occurred in 92 (28%) patients in the avelumab maintenance group, 118 (36%) in the avelumab combination group, and 64 (19%) in the control group. Treatment-related deaths occurred in one (<1%) patient in the avelumab maintenance group (due to atrial fibrillation) and one (<1%) patient in the avelumab combination group (due to disease progression).

Interpretation Although no new safety signals were observed, results do not support the use of avelumab in the frontline treatment setting. Alternative treatment regimens are needed to improve outcomes in patients with advanced epithelial ovarian cancer.

Lancet Oncol 2021; 22: 1275–89

Published Online
August 4, 2021
[https://doi.org/10.1016/S1470-2045\(21\)00342-9](https://doi.org/10.1016/S1470-2045(21)00342-9)

*Co-international leads and contributed equally to the manuscript

Arizona Oncology (US Oncology Network), Phoenix, AZ, USA (Prof B J Monk MD); Department of Obstetrics and Gynecology, University of Arizona College of Medicine, Phoenix, AZ, USA (Prof B J Monk, S M Bhoola MD); Department of Obstetrics and Gynecology, Creighton University School of Medicine at Dignity Health St Joseph's Hospital and Medical Center, Phoenix, AZ, USA (Prof B J Monk); University of Milan-Bicocca, Milan, Italy (N Colombo MD); Istituto Europeo di Oncologia, IRCCS, Milan, Italy (N Colombo); Princess Margaret Cancer Centre, Toronto, ON, Canada (Prof A M Oza MD); Saitama Medical University International Medical Center, Hidaka-City, Saitama, Japan (Prof K Fujiwara MD); Winthrop P Rockefeller Cancer Institute, Little Rock, AR, USA (Prof M J Birrer MD); Virginia Commonwealth University, Massey Cancer Center, Richmond, VA, USA (Prof L Randall MD); I M Sechenov First Moscow State Medical University, Moscow, Russia (E V Poddubskaya MD); Clinical Center Vitamed, Moscow, Russia (E V Poddubskaya); Gynecologic Oncology Unit, Fondazione Policlinico Universitario Agostino Gemelli

IRCCS, Rome, Italy
(Prof G Scambia MD); Lviv State
Oncological Regional
Treatment and Diagnostic
Center, Lviv, Ukraine
(Y V Shparyk MD); Research
Institute and Hospital,
National Cancer Center,
Goyang, South Korea
(M C Lim MD); Arizona
Oncology Associates PC-HAL,
Tempe, AZ, USA (S M Bhoola);
Gynecologic Oncology, Cancer
and Blood Specialists of
Arizona, Gilbert, AZ, USA
(S M Bhoola); Severance
Hospital, Yonsei University
Health System, Seoul,
South Korea (Prof J Sohn MD);
Department of Breast and
Medical Oncology, National
Cancer Center Hospital,
Chuo-ku, Tokyo, Japan
(K Yonemori MD); Pfizer
Oncology, Pfizer, San Diego,
CA, USA (R A Stewart PhD);
Oncology Research and
Development, AstraZeneca,
Cambridge, UK (R A Stewart);
Pfizer Oncology, Pfizer,
New York, NY, USA
(X Zhang PhD,
J Perkins Smith MD); Global
Product Development, Pfizer,
Taipei, Taiwan (C Linn MD);
UCL Cancer Institute and UCL
Hospitals, London, UK
(Prof J A Ledermann MD)

Correspondence to:
Prof Bradley J Monk, Arizona
Oncology, Biltmore Cancer
Center, Phoenix, AZ 85016, USA
bradley.monk@usoncology.
com

Funding Pfizer and Merck KGaA, Darmstadt, Germany.

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Introduction

Ovarian cancer is responsible for approximately 185 000 deaths worldwide each year.¹ Generally more than 70% of women diagnosed with ovarian cancer have advanced disease.^{2,3} Ovarian cancer is a heterogenous disease; however, most tumours (approximately 90%) are epithelial ovarian cancers,³ which include cancers originating from cells lining the fallopian tubes and peritoneum, which are managed similarly.⁴ Current standard-of-care frontline treatment for patients with advanced epithelial ovarian cancers consists of combination carboplatin and paclitaxel chemotherapy before or after debulking surgery.^{3,4} Frontline treatment with bevacizumab, an anti-VEGF antibody, in combination with chemotherapy and followed by bevacizumab maintenance is approved in various countries worldwide for the treatment of advanced epithelial ovarian cancer irrespective of biomarker status. The addition of bevacizumab has been associated with an improvement in progression-free survival in overall trial populations; however, only women at high risk of disease recurrence had an overall survival benefit.^{5,6} Trials including combination chemotherapy regimens have shown that most patients with ovarian cancer (around 70–80%) should respond to initial treatment, but median progression-free survival ranges from 10 to 17 months,^{5,7–10} and approximately 70% of patients relapse within 3 years.³

Although not considered a strongly immunogenic cancer, epithelial ovarian tumours are characterised by immunological activity. PD-L1, a key suppressor of T cell function, is expressed on ovarian tumour cells and tumour-infiltrating lymphocytes in more than half of patients.¹¹ The presence of tumour-infiltrating lymphocytes, specifically CD8-positive T cells, is associated with longer overall survival.¹² Trials of immune checkpoint inhibitors as monotherapy in patients with previously treated epithelial ovarian cancer have shown modest clinical activity,^{13–15} prompting interest in the use of immune checkpoint inhibitors in combination with established treatments. Furthermore, accumulating evidence suggests that chemotherapy drugs might regulate antitumour immune responses. For example, cytotoxic chemotherapy can stimulate tumour immunosurveillance by increasing antigen release and presentation, modifying the suppressive tumour microenvironment (eg, by suppressing T regulatory cells), and promoting infiltration of T cells, potentially increasing tumour susceptibility to immune attack.^{16,17} Additionally, a hypothesis is that DNA-damaging chemotherapies might increase the mutational burden within tumours, thereby increasing the repertoire of neoantigens available for tumour-directed immune responses.¹⁸ Therefore, combining chemotherapy with an immune checkpoint inhibitor could provide improved clinical activity by stimulating the activity of CD8-positive

Research in context

Evidence before this study

We did a literature search on PubMed on May 4, 2021, using the terms (“ovarian cancer” OR “epithelial ovarian cancer”) AND (“PD-1” OR “PD-L1” OR “programmed death” OR “checkpoint inhibitor”) AND (“study” OR “trial”) for clinical trials of immune checkpoint inhibitors in epithelial ovarian cancer published at any time from database inception in English. We identified 15 manuscripts reporting data from phase 1–3 trials in various epithelial ovarian cancer populations (five phase 1 trials, one phase 1–2 trial, six phase 2 trials, and one phase 3 trial). No manuscripts were found that reported a clinical study of an immune checkpoint inhibitor as maintenance treatment in patients with epithelial ovarian cancer. One manuscript reported a phase 3 clinical study of an immune checkpoint inhibitor combined with chemotherapy in the frontline setting; the study (IMagyn050) investigated the addition of atezolizumab, an anti-PD-L1 antibody, versus the addition of placebo to platinum-based chemotherapy and bevacizumab in patients with treatment-naïve stage III–IV epithelial ovarian cancer. In IMagyn050, the addition of atezolizumab did not significantly improve progression-free survival in the overall or PD-L1-positive populations.

Added value of this study

To our knowledge, JAVELIN Ovarian 100 is one of the first phase 3 trials of an immune checkpoint inhibitor in patients with previously untreated epithelial ovarian cancer to be reported. The trial did not meet either of its two primary objectives of significantly improving progression-free survival with chemotherapy followed by avelumab maintenance or chemotherapy plus avelumab followed by avelumab maintenance versus chemotherapy followed by observation, and was terminated at the interim analysis in accordance with predefined futility boundaries for both avelumab groups. No new safety signals were observed in either avelumab group.

Implications of all the available evidence

The findings from this trial suggest that the addition of an immune checkpoint inhibitor to frontline chemotherapy does not improve progression-free survival in the overall population, highlighting that further study is needed to determine whether immune checkpoint inhibitors have a role in the frontline treatment of epithelial ovarian cancer.

tumour-infiltrating lymphocytes and overcoming immunosuppressive mechanisms. Increased clinical activity and manageable toxicity have been reported in trials of combined chemotherapy and immunotherapy regimens in other tumour types;^{19–21} however, data in epithelial ovarian cancer with similar treatment regimens are scarce.

Avelumab, an anti-PD-L1 monoclonal antibody, showed clinical activity as monotherapy in a phase 1b cohort that included 125 heavily treated patients with resistant or refractory epithelial ovarian cancer, with an objective response in 12 (10%) of the 125 patients.¹³ However, in the phase 3 JAVELIN Ovarian 200 trial, avelumab did not show superiority over pegylated liposomal doxorubicin (PLD), either as monotherapy or in combination with PLD in patients with platinum-resistant or platinum-refractory epithelial ovarian cancer.²²

In this Article, we report results from the randomised, open-label, phase 3 JAVELIN Ovarian 100 trial, which compared avelumab in combination with platinum-based chemotherapy followed by avelumab maintenance or chemotherapy followed by avelumab maintenance versus chemotherapy alone (followed by observation) in patients with treatment-naïve epithelial ovarian cancer.

Methods

Study design and participants

JAVELIN Ovarian 100 was a global, open-label, three-arm, parallel, randomised, phase 3 trial run at 159 hospitals and cancer treatment centres in 25 countries (Bulgaria, Canada, Croatia, Estonia, Germany, Hong Kong, Hungary, Ireland, Italy, Japan, Latvia, Mexico, the Netherlands, Poland, Romania, Russia, Singapore, Slovakia, South Korea, Switzerland, Taiwan, Turkey, Ukraine, the UK, and the USA; appendix pp 2–8). Eligible women were aged 18 years and older (≥ 20 years in Japan); had previously untreated, histologically confirmed, stage III–IV epithelial ovarian, fallopian tube, or primary peritoneal cancer (including malignant mixed Müllerian tumours with a high-grade serous component); were candidates for platinum-based chemotherapy; and had received debulking surgery or were candidates for neoadjuvant chemotherapy. Staging was per the American Joint Committee on Cancer and Union for International Cancer Control TNM cancer staging system and the International Federation of Gynaecology and Obstetrics staging system (2014 edition). Patients enrolled before initiation of neoadjuvant chemotherapy underwent interval debulking surgery after three cycles of chemotherapy (with or without avelumab) and then completed the remaining three cycles post-surgery. Other eligibility criteria included an Eastern Cooperative Oncology Group performance status of 0 or 1; estimated life expectancy of at least 12 weeks; adequate haematological function (absolute neutrophil count $\geq 1.5 \times 10^9$ cells per L, haemoglobin ≥ 9 g/dL, and platelet count $\geq 100 \times 10^9$ per L), hepatic function (aspartate aminotransferase and alanine aminotransferase concentrations

≤ 2.5 times the upper limit of normal and total bilirubin concentration ≤ 1.5 times the upper limit of normal), and renal function (creatinine clearance ≥ 50 mL/min calculated according to the Cockcroft-Gault equation); and a negative pregnancy test and use of effective contraception (women of childbearing potential). All patients were required to have an archival formalin-fixed, paraffin-embedded tumour tissue block or a minimum of 15 tumour slides. If archived tissue was not available, a de-novo (ie, fresh) tumour sample was obtained. Exclusion criteria included non-epithelial tumour or tumour with low malignant potential (ie, borderline tumour); mucinous tumours; cancer for which bevacizumab was identified as a clinically beneficial frontline treatment; planned intraperitoneal chemotherapy; previous treatment with a T-cell-targeting immune checkpoint inhibitor; known brain, leptomeningeal, or spinal cord metastases; other cancer diagnosis within 5 years; known hypersensitivity to monoclonal antibodies, carboplatin, paclitaxel, or other platinum-containing compounds (grade ≥ 3 reaction per the US National Cancer Institute's Common Terminology Criteria for Adverse Events [CTCAE] version 4.03); and serious cardiovascular disease or other severe medical condition. Full eligibility criteria are listed in the protocol (appendix).

The trial was done in accordance with the ethics principles of the Declaration of Helsinki and the International Council on Harmonisation guidelines on Good Clinical Practice. The protocol was approved by the institutional review board or ethics committee of each centre. All patients (or their legal representatives) provided written informed consent before enrolment.

Randomisation and masking

Patients were enrolled and centrally randomly assigned (1:1:1) by study investigators via interactive response technology to receive either chemotherapy (carboplatin plus paclitaxel) followed by avelumab maintenance (referred to as the avelumab maintenance group), chemotherapy plus avelumab followed by avelumab maintenance (referred to as the avelumab combination group), or chemotherapy followed by observation (control group), by stratified permuted block randomisation with a block size of six. Randomisation was stratified by paclitaxel regimen (once a week vs once every 3 weeks) and resection status (residual tumour; complete or microscopic vs incomplete ≤ 1 cm vs incomplete > 1 cm vs neoadjuvant). The trial was open label, although patients and investigators were masked to assignment to the two chemotherapy groups without avelumab at time of randomisation until completion of the chemotherapy phase. Personnel delivering the chemotherapy were also masked to treatment assignment. The sponsor and an independent central review committee (third party) remained masked to treatment assignments until after study termination.

See Online for appendix

Procedures

Avelumab was administered at a dose of 10 mg/kg by 1-h intravenous infusion once every 3 weeks in the chemotherapy phase and 10 mg/kg once every 2 weeks in the maintenance phase. Chemotherapy regimens consisted of six cycles of paclitaxel at either 80 mg/m² by 1-h intravenous infusion once weekly or 175 mg/m² by 3-h intravenous infusion once every 3 weeks (investigators' choice; once selected, dose was not changed for the study duration) plus a carboplatin area under the serum-concentration-time curve of 5 or 6 (according to the Calvert formula) by 1-h intravenous infusion once every 3 weeks. Carboplatin was administered 1 h after completing the paclitaxel infusion. Premedication with an antihistamine (eg, oral or intravenous diphenhydramine 25–50 mg or equivalent) and paracetamol (oral or intravenous, 500–650 mg or equivalent) was mandatory 30–60 min before each avelumab infusion. Premedications to minimise toxicities related to chemotherapy were administered according to local practice guidelines. For avelumab, no dose reductions were permitted, although doses could be omitted for persisting toxicity on the basis of investigator judgement; the carboplatin dose or paclitaxel dose (or both) could be reduced following clinically significant toxicity on the basis of investigator judgement. Treatment with a non-zero dose that was less than 90% of the planned dose was classified as a dose reduction.

After completing the chemotherapy phase, if the patient had stable disease or a partial or complete response, they entered the maintenance phase. Treatments or observation were continued until disease progression (assessed by an investigator but confirmed by blinded independent central review), unacceptable toxicity, or withdrawal (potential reasons leading to withdrawal included global deterioration of health status, pregnancy, substantial protocol deviation, patient refusal, loss to follow-up, termination of the study by the sponsor, or death; as listed in the protocol [appendix]); avelumab could be continued while awaiting confirmation of disease progression according to the investigator's clinical judgment. Treatment in the maintenance phase was received for a maximum of 24 months (excluding the chemotherapy phase). Patients who discontinued (or, if receiving observation maintenance, reached the end of the maintenance phase or withdrew) were followed up every 12 weeks until death or the end of the study. Crossover between study groups was not permitted.

Tumours were assessed by CT or MRI at baseline, after three cycles of chemotherapy, and at completion of chemotherapy to determine eligibility for maintenance. Patients who underwent interval debulking surgery were required to have an additional tumour assessment after surgery. Assessments were done every 12 weeks in the maintenance phase until confirmed disease progression, irrespective of subsequent anticancer therapy. Objective

tumour response was evaluated by blinded independent central review according to the Response Evaluation Criteria in Solid Tumors (RECIST) version 1.1. Complete and partial responses and progressive disease were confirmed by repeated imaging up to 4 weeks after initial documentation. Blood samples were taken at each treatment visit for haematology. Samples were also taken on day 1 of each cycle in the chemotherapy phase and once every two weeks in the maintenance phase for other routine laboratory analyses, including core serum chemistry and haemostaseology. Urine samples were taken at screening and on day 1 of each cycle in the maintenance phase for urinalysis. Adrenocorticotrophic hormone, free thyroxine, and thyroid-stimulating hormone concentrations were tested at screening, on day 1 of each odd cycle of the chemotherapy phase, day 1 of the third cycle of the maintenance phase, and then every 12 weeks thereafter while on treatment. CA-125 was assessed locally by investigators at screening, on day 1 of each cycle in both the chemotherapy and maintenance phases, and at the end of treatment. The full schedule of assessments is provided in the protocol (appendix). Adverse events (recorded at each visit) and laboratory abnormalities were graded by investigators according to the CTCAE version 4.03. Immune-related adverse events and infusion-related reactions were identified by investigators with a prespecified list of terms in the Medical Dictionary for Regulatory Activities at each visit, in addition to comprehensive medical review by employees of the sponsor.

We assessed PD-L1 expression in pretreatment tissue samples using an immunohistochemical assay based on the SP263 antibody (Ventana Medical Systems, Tuscon, AZ, USA). Selection of the PD-L1 cutoff was based on post-hoc analyses of several scoring algorithms and cutoff values from the JAVELIN Ovarian 200 trial,²² and the optimal cutoff for predicting improved clinical activity for the combination of avelumab and chemotherapy was selected. A sample was considered PD-L1-positive if the percentage of tumour cells expressing membranous PD-L1 was 1% or greater, the percentage of tumour area populated by PD-L1-positive immune cells was 5% or greater, or both.

An external data monitoring committee was established to review safety and efficacy data from the trial.

Outcomes

The primary endpoint was progression-free survival assessed by blinded independent central review (defined as the time from randomisation to the date of the first documented disease progression per RECIST 1.1 or death due to any cause, whichever occurred first). Secondary endpoints were overall survival (defined as the time from randomisation to the date of death due to any cause); progression-free survival determined by investigator assessment per RECIST 1.1; maintenance progression-free survival (for patients who did not have

disease progression as per blinded independent central review and investigator assessment during the chemotherapy phase and entered the maintenance phase; defined as the time from initiating maintenance treatment to the date of first documented disease progression per RECIST 1.1 or death due to any cause, whichever occurred first); objective response (defined as a complete response or partial response per RECIST 1.1) and duration of response (defined for patients with an objective response as the time from the first documentation of objective response to the first documentation of disease progression or death due to any cause, whichever occurs first), by blinded independent central review and investigator assessment; pathological complete response (defined as a chemotherapy response score of 3 according to Bohm et al²³ for patients who received study treatment as neoadjuvant therapy for three cycles and underwent interval debulking surgery), progression-free survival 2 (defined as the time from randomisation to the start of second subsequent treatment after objective disease progression or death due to any cause, whichever occurred first), and progression-free survival according to Gynecological Cancer Intergroup criteria (all by investigator assessment); safety and tolerability (based on adverse events, laboratory abnormalities, vital signs, and ECG); pharmacokinetic parameters; immunogenicity of avelumab; tumour biomarker assessments (including, but not limited to, PD-L1 expression); and patient-reported outcomes. Progression-free survival 2 and progression-free survival defined by Gynecological Cancer Intergroup criteria are not reported in this Article because the required assessments were not completed after early termination of the trial. Pharmacokinetic parameters, immunogenicity, and patient-reported outcomes are not reported because this Article focuses on the clinical aspects of the trial and because these analyses had limited relevance given that the trial did not meet its primary endpoint. Additional biomarker analyses except for PD-L1 expression are ongoing and not presented in this Article.

Statistical analysis

The trial aimed to show superiority of avelumab in combination with and following chemotherapy, or following chemotherapy alone, in prolonging progression-free survival, compared with the control regimen of chemotherapy followed by observation, in all randomly assigned patients (analysed by intention to treat). Two independent and adequately powered comparisons of progression-free survival were done: avelumab maintenance versus the control group, and avelumab combination versus the control group. The study used a two-look group-sequential design with a Lan-DeMets (O'Brien-Fleming) α -spending function to determine the efficacy boundary and a γ -family β -spending function to determine the non-binding futility boundary, with one planned interim analysis and one final analysis based on the primary endpoint. The overall type I error rate was

maintained at or lower than a one-sided significance level of 0.025 by allocating an α level of 0.0125 to both progression-free survival comparisons; a fraction of α (0.0022) for efficacy was allocated to the interim analysis. For each progression-free survival comparison, we estimated that 272 events within each comparison would provide the trial with 90% power to detect a hazard ratio (HR) of 0.65 using a one-sided log-rank test. An interim analysis was planned when approximately 181 (67%) of 272 events had occurred for each progression-free survival comparison. We planned that if the value of the test statistic exceeded the efficacy boundary ($z < -2.848$, $p < 0.002$), then the experimental treatment (or the study) could be stopped for efficacy. We also planned that if the value of the test statistic exceeded the futility boundary ($z > -0.804$, $p > 0.211$), then the experimental treatment (or the study) could be stopped for futility.²⁴ For overall survival comparisons, 376 events within each comparison would provide 70% power to detect a HR of 0.75. These comparisons were not done because the early termination of the trial meant that the overall survival analysis was immature. This Article reports efficacy results from the interim analysis (data cutoff Sept 7, 2018) and updated safety data and prespecified analyses based on baseline PD-L1 status from an additional later cutoff (May 16, 2019). Immune-related adverse events were not updated as they required sponsor adjudication, and most immune-related adverse events occurred in the first few months of receiving avelumab.

All efficacy outcomes were analysed in all patients who were randomly assigned to study treatment, and safety was analysed in all patients who received at least one dose of study treatment. Time-to-event endpoints were estimated with the Kaplan-Meier method, and 95% CIs for the median were calculated by the Brookmeyer-Crowley method. Duration of follow-up was estimated with the reverse Kaplan-Meier method.²⁵ The Cox proportional hazards model was used to calculate HRs and corresponding 95% CIs for progression-free survival (including prespecified exploratory subgroup analyses) and overall survival. Subgroup analyses were done by randomisation stratification factors, age (≥ 65 vs < 65 years), race (Asian vs Black or African American vs other vs White), ethnicity (Hispanic or Latino vs non-Hispanic or non-Latino), pooled geographical region (North America vs Asia vs rest of the world vs Europe), *BRCA1/2* mutation status (negative vs unknown vs positive), disease stage (IV vs III), Eastern Cooperative Oncology Group performance status (≥ 1 vs 0), baseline CA-125 (> 2 times upper limit of normal vs ≤ 2 times upper limit of normal; as a reference standard²⁶), and baseline PD-L1 status (negative vs positive). Progression-free survival and overall survival were analysed with a one-sided stratified log-rank test. Objective response rates and rates of pathological complete response were calculated for each treatment group, along with two-sided 95% CIs (Clopper-Pearson method).

Statistical analyses were done in SAS (version 9.4). This study is registered with ClinicalTrials.gov, NCT02718417.

Role of the funding source

The trial was sponsored by Pfizer as part of an alliance between Pfizer and Merck KGaA, Darmstadt, Germany. The sponsor provided the study drugs, worked with a protocol steering committee to design the trial and to collect, analyse, and interpret the data, and provided funding for a professional medical writer who had access to the data.

Results

Between May 19, 2016, and Jan 23, 2018, 998 patients were enrolled and randomly assigned. 332 patients were allocated to the avelumab maintenance group, 331 to the avelumab combination group, and 335 to the control group (figure 1). Baseline characteristics were balanced across the three groups (table 1). At the planned interim analysis (data cutoff Sept 7, 2018), test statistics for progression-free survival in both avelumab groups had crossed the prespecified futility boundaries. The trial was stopped due to futility of efficacy in alignment with the recommendation of both the independent data monitoring committee and the protocol steering committee. In the chemotherapy phase, as of the updated safety data cutoff (May 16, 2019), at least one dose of assigned treatment was received by 328 patients in the avelumab maintenance group, 329 patients in the avelumab combination group, and 334 patients in the control group. Among treated patients, nearly all had received premedication with systemic corticosteroids (328 [100%] patients in the avelumab maintenance group, 327 [99%] of 329 in the avelumab combination group, and 333 [>99%] of 334 in the control group).

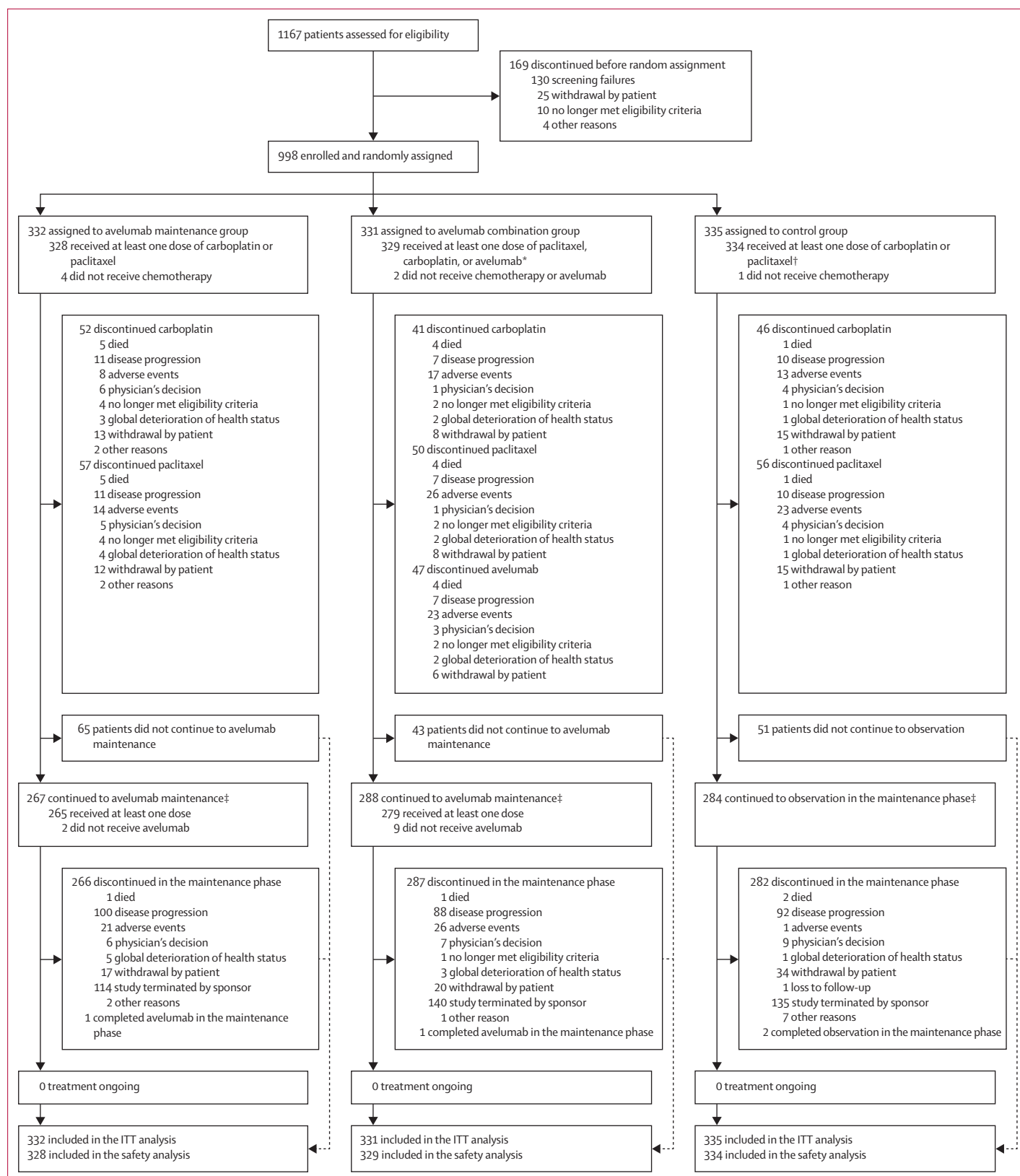
In the analysis of all randomly assigned patients, median duration of follow-up for progression-free survival was 10.8 months (IQR 7.1–14.9) for all patients; by group, this was 11.1 months (7.0–15.3) for the avelumab maintenance group, 11.0 months (7.4–14.5) for the avelumab combination group, and 10.2 months (6.7–14.0) for the control group. Median duration of treatment for all study drugs in the chemotherapy phase was 19 weeks or longer. Most randomly assigned patients (>80%) completed the chemotherapy phase and entered the maintenance phase (figure 1). Among patients assigned to the avelumab maintenance group, 275 (83%) and 280 (84%) completed the paclitaxel and carboplatin treatments, respectively; among those assigned to the avelumab combination group, 284 (86%), 281 (85%), and 290 (88%) completed the avelumab, paclitaxel, and carboplatin treatments, respectively; and among those assigned to the control group, 279 (83%) and 289 (86%) completed the paclitaxel and carboplatin treatments, respectively. The most common reasons for treatment discontinuation during the chemotherapy

phase were adverse events, withdrawal by the patient, and progressive disease (figure 1). In the maintenance phase, 265 (99%) of 267 patients who continued to maintenance in the avelumab maintenance group received at least one dose of avelumab; this proportion was 279 (97%) of 288 in the avelumab combination group. Median duration of avelumab treatment was 35.7 weeks (IQR 21.9–52.0) in the avelumab maintenance group and 36.0 weeks (23.9–52.9) in the avelumab combination group. The most common reason for discontinuation in all three groups in the maintenance phase was study termination (114 [34%] of 332 randomly assigned patients in the avelumab maintenance group, 140 [42%] of 331 in the avelumab combination group, and 135 [40%] of 335 in the control group), and as of May 16, 2019, no patient remained on the study.

Analysis of all efficacy endpoints was based on blinded independent central review unless otherwise specified. As of the interim analysis cutoff (Sept 7, 2018), disease progression or death had occurred in 99 (30%) of 332 patients in the avelumab maintenance group, 88 (27%) of 331 patients in the avelumab combination group, and 70 (21%) of 335 patients in the control group (intention-to-treat population). Median progression-free survival was 16.8 months (95% CI 13.5–not estimable [NE]) in the avelumab maintenance group, 18.1 months (14.8–NE) in the avelumab combination group, and NE (18.2 months–NE) in the control group (figure 2). The stratified HR for progression-free survival was 1.43 (95% CI 1.05–1.95; one-sided $p=0.99$) with the avelumab maintenance regimen and 1.14 (0.83–1.56; one-sided $p=0.79$) with the avelumab combination regimen, versus control treatment. Prespecified exploratory subgroup analyses of progression-free survival based on patient and disease characteristics showed similar results (figure 3). Median progression-free survival by investigator assessment was 13.8 months (95% CI 12.1–15.9) in the avelumab maintenance group, 16.1 months (13.9–19.4) in the avelumab combination group, and 15.0 months (13.2–18.7) in the control group. The stratified HR for progression-free survival determined by investigator assessment was 1.21 (95% CI 0.94–1.58; one-sided $p=0.93$) with the avelumab maintenance regimen and 0.90 (0.69–1.19; one-sided $p=0.24$) with the avelumab combination regimen, versus control treatment. Maintenance progression-free survival was assessed in patients

Figure 1: Trial profile

The profile is from the updated safety data cutoff of May 16, 2019. Because both avelumab groups had crossed the prespecified futility boundaries, the trial was stopped due to futility of efficacy in alignment with the recommendation of both the independent data monitoring committee and the protocol steering committee. ITT=intention to treat. *327 patients received at least one dose of carboplatin, 329 received at least one dose of paclitaxel, and 327 received at least one dose of avelumab. †331 patients received at least one dose of carboplatin and 333 received at least one dose of paclitaxel. ‡Patients who discontinued the earlier drugs were eligible to enter the maintenance phase.



	Chemotherapy followed by avelumab (n=332)	Chemotherapy plus avelumab followed by avelumab (n=331)	Chemotherapy followed by observation (n=335)
Median age (IQR), years	59.0 (52.0–67.0)	60.0 (50.0–66.0)	57.0 (49.0–66.0)
Eastern Cooperative Oncology Group performance status*			
0	186 (56%)	179 (54%)	196 (59%)
1	145 (44%)	150 (45%)	136 (41%)
2	0	2 (1%)	2 (1%)
Pooled geographic region			
North America	93 (28%)	103 (31%)	85 (25%)
Europe	147 (44%)	139 (42%)	149 (44%)
Asia	78 (23%)	79 (24%)	90 (27%)
Rest of the world	14 (4%)	10 (3%)	11 (3%)
Race			
White	236 (71%)	238 (72%)	236 (70%)
Asian	86 (26%)	82 (25%)	95 (28%)
Black or African American	2 (1%)	4 (1%)	1 (<1%)
Other†	8 (2%)	7 (2%)	3 (1%)
Site of primary tumour‡			
Ovary	261 (79%)	259 (78%)	270 (81%)
Peritoneum	38 (11%)	41 (12%)	32 (10%)
Fallopian tube	33 (10%)	31 (9%)	32 (10%)
Not reported	0	1 (<1%)	1 (<1%)
Histology			
High-grade serous	258 (78%)	257 (78%)	247 (74%)
Low-grade serous	18 (5%)	23 (7%)	21 (6%)
Clear cell	19 (6%)	15 (5%)	21 (6%)
Endometrioid	12 (4%)	10 (3%)	10 (3%)
Other epithelial ovarian cancer§	25 (8%)	26 (8%)	36 (11%)
Histopathological grade			
Grade 1	16 (5%)	16 (5%)	22 (7%)
Grade 2	34 (10%)	36 (11%)	31 (9%)
Grade 3	278 (84%)	265 (80%)	272 (81%)
Not reported	4 (1%)	14 (4%)	10 (3%)
Disease stage at diagnosis			
III	37 (11%)	32 (10%)	38 (11%)
IIIB	25 (8%)	32 (10%)	31 (9%)
IIIC	190 (57%)	201 (61%)	178 (53%)
IV	46 (14%)	35 (11%)	44 (13%)
IVA	11 (3%)	12 (4%)	10 (3%)
IVB	20 (6%)	17 (5%)	27 (8%)
Unknown¶	3 (1%)	3 (1%)	7 (2%)

(Table 1 continues in next column)

who had not had disease progression in the chemotherapy phase and subsequently entered the maintenance phase; this population (by blinded independent central review) comprised 248 patients in the avelumab maintenance group, 267 patients in the avelumab combination group,

	Chemotherapy followed by avelumab (n=332)	Chemotherapy plus avelumab followed by avelumab (n=331)	Chemotherapy followed by observation (n=335)
(Continued from previous column)			
Measurable disease (blinded independent central review)			
Yes	232 (70%)	228 (69%)	226 (67%)
No	100 (30%)	102 (31%)	109 (33%)
No disease	0	1 (<1%)	0
Paclitaxel regimen			
Once a week	100 (30%)	102 (31%)	104 (31%)
Once every 3 weeks	232 (70%)	229 (69%)	231 (69%)
Resection (residual tumour)			
Complete resection or microscopic disease	105 (32%)	105 (32%)	105 (31%)
Incomplete resection ≤1 cm	34 (10%)	35 (11%)	36 (11%)
Incomplete resection >1 cm	57 (17%)	55 (17%)	57 (17%)
Neoadjuvant	136 (41%)	136 (41%)	137 (41%)
PD-L1 status**			
Positive	158 (48%)	160 (48%)	169 (50%)
Negative	112 (34%)	103 (31%)	111 (33%)
Not evaluable	62 (19%)	68 (21%)	55 (16%)
CA-125 expression			
≤2 × ULN	86 (26%)	73 (22%)	93 (28%)
>2 × ULN	238 (72%)	255 (77%)	236 (70%)
Not reported	8 (2%)	3 (1%)	6 (2%)

Data are n (%) unless otherwise specified. ULN=upper limit of normal. *Not reported for two patients at baseline (scores were available at screening and randomisation; one patient in the chemotherapy followed by avelumab group, one patient in the chemotherapy followed by observation group). †Includes American Indian or Alaska Native (one patient in the chemotherapy followed by avelumab group) and other (remaining patients). ‡One patient in the chemotherapy plus avelumab followed by avelumab group had a primary tumour recorded in two sites (ovary and peritoneum); primary tumour site not recorded but eligibility confirmed in two patients. §Includes adenocarcinoma, undifferentiated carcinoma, and not reported. ¶Includes unknown stage, staging not applicable, or stage not reported; eligibility confirmed at screening. ||Recorded at randomisation. **PD-L1-positive status was defined as expression in at least 1% of tumour cells, at least 5% of immune cells, or both (Ventana PD-L1 SP263 immunohistochemistry assay).

Table 1: Baseline characteristics

and 247 patients in the control group. Median maintenance progression-free survival was 13.6 months (95% CI 9.3–NE), 13.8 months (11.1–NE), and NE (13.8–NE), respectively. The stratified HR for maintenance progression-free survival was 1.56 (95% CI 1.08–2.27; one-sided $p=0.99$) with avelumab maintenance and 1.26 (0.86–1.85; one-sided $p=0.89$) with avelumab combination treatment, versus control treatment. By investigator assessment, the population assessed for maintenance progression-free survival comprised 252 patients in the avelumab maintenance

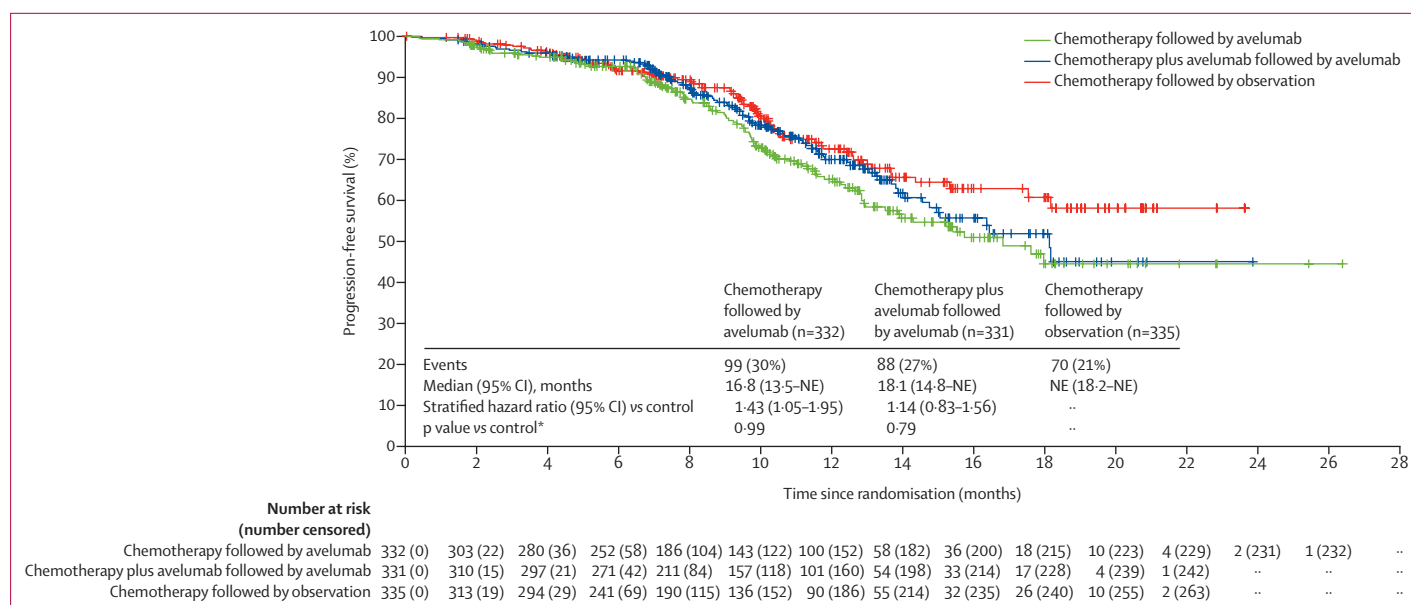


Figure 2: Progression-free survival

The data cutoff for analysis was Sept 7, 2018. NE=not estimable. *One-sided stratified log-rank test.

group, 271 patients in the avelumab combination group, and 252 patients in the control group. Median maintenance progression-free survival by investigator assessment was 10.4 months (95% CI 8.2–13.6), 11.6 months (9.9–13.8), and 12.7 months (9.5–NE), respectively. The stratified HR was 1.27 (95% CI 0.94–1.71; one-sided $p=0.94$) with avelumab maintenance and 0.99 (0.73–1.36; one-sided $p=0.48$) with avelumab combination treatment, versus control treatment.

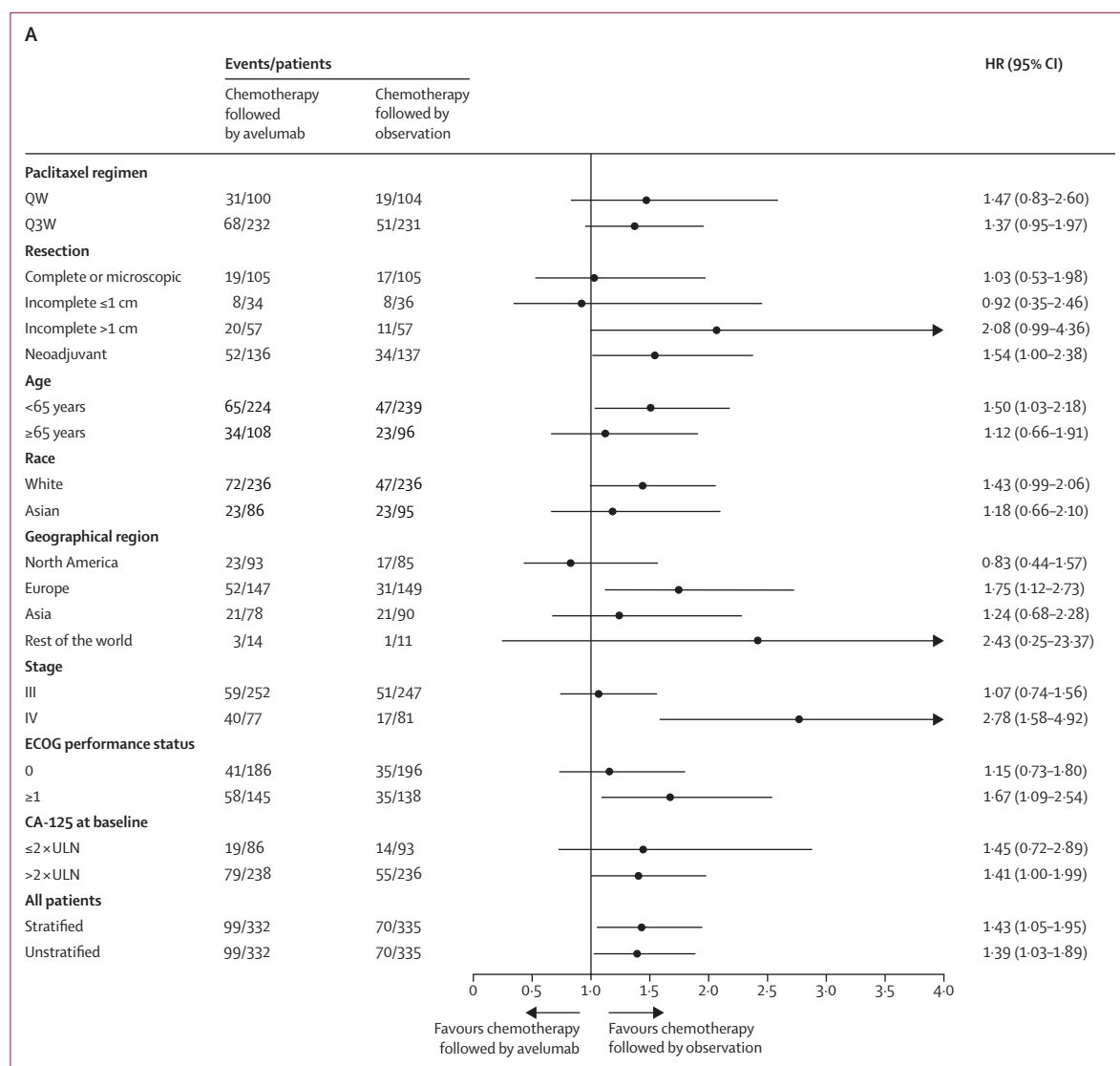
Overall survival data were not mature at the time of the interim analysis, with a total of 54 deaths across the three groups (20 [6%] in the avelumab maintenance group, 21 [6%] in the avelumab combination group, and 13 [4%] in the control group; appendix p 23). Median follow-up for overall survival was 12.4 months (IQR 9.0–15.9) for all patients: 12.6 months (9.1–16.0) in the avelumab maintenance group, 12.6 months (9.5–16.1) in the avelumab combination group, and 11.8 months (8.5–15.6) in the control group. Objective response and response duration data are also summarised in the appendix (pp 9–12). Interval debulking surgery after neoadjuvant treatment (3 cycles of assigned chemotherapy) was received by 108 patients in the avelumab maintenance group, 115 patients in the avelumab combination group, and 116 patients in the control group. Among these patients, the number with pathological complete response was 17 (16% [95% CI 12–29]) in the avelumab maintenance group, 20 (17% [13–30]) in the avelumab combination group, and 30 (26% [21–40]) in the control group.

In a prespecified analysis (data cutoff May 16, 2019), the predictive role of tumour PD-L1 status was assessed in 813 evaluable patients. For PD-L1-positive and

PD-L1-negative subgroups, progression-free survival by blinded independent central review and by investigator assessment is shown in the appendix (pp 24–27), although the data were inconclusive.

As of May 16, 2019, treatment-emergent adverse events of any grade or causality occurred in 323 (98%) of 328 patients in the avelumab maintenance group, 328 (>99%) of 329 patients in the avelumab combination group, and 321 (96%) of 334 patients in the control group (table 2 and appendix pp 14–21). Grade 3–5 adverse events occurred in 223 (68%) patients in the avelumab maintenance group, 238 (72%) patients in the avelumab combination group, and 210 (63%) patients in the control group. The most common adverse events of grade 3 or worse (in $\geq 10\%$ of patients in all groups) were anaemia (69 [21%] patients in the avelumab maintenance group, 63 [19%] in the avelumab combination group, and 53 [16%] in the control group), neutropenia (91 [28%], 99 [30%], and 88 [26%], respectively), and neutrophil count decrease (49 [15%], 45 [14%], and 59 [18%]). Serious adverse events of any grade occurred in 92 (28%) patients in the avelumab maintenance group, 118 (36%) patients in the avelumab combination group, and 64 (19%) patients in the control group. Grade 3–5 serious adverse events occurred in 72 (22%), 93 (28%), and 48 (14%) patients, respectively.

In the avelumab maintenance, avelumab combination, and control groups, adverse events led to discontinuation of any study drug in 42 (13%), 63 (19%), and 24 (7%) patients, respectively. Adverse events resulted in death in five (2%) patients in the avelumab maintenance group (pulmonary embolism [$n=2$], disease progression [$n=1$], atrial fibrillation [$n=1$], and embolism [$n=1$]), six (2%)



(Figure 3 continues on next page)

patients in the avelumab combination group (disease progression [n=3], multiple organ dysfunction syndrome [n=1], cardiopulmonary failure [n=1], and perforation and abdominal abscess [n=1; in a patient who also had grade 3 small intestinal obstruction]), and three (1%) patients in the control group (death from unspecified cause [n=1], malignant neoplasm progression [n=1], and pulmonary embolism [n=1]; appendix pp 14–21). Dose reductions are detailed in the appendix (p 13).

Treatment-related adverse events of any grade occurred in 315 (96%) patients in the avelumab maintenance group, 324 (98%) patients in the avelumab combination group, and 318 (95%) patients in the control group. Grade 3–5 treatment-related adverse events occurred in 175 (53%), 205 (62%), and 186 (56%) patients, respectively. In the avelumab maintenance, avelumab combination,

and control groups, serious treatment-related adverse events occurred in 43 (13%), 62 (19%), and 29 (9%) patients, respectively; the most common (in ≥2% patients in any group) were febrile neutropenia (ten [3%] patients), anaemia (five [2%]), and vomiting (five [2%]) in the avelumab maintenance group, febrile neutropenia (seven [2%]), anaemia (seven [2%]), vomiting (seven [2%]), thrombocytopenia (five [2%]), and nausea (five [2%]) in the avelumab combination group, and febrile neutropenia (seven [2%]) in the control group.

Treatment-related adverse events led to discontinuation of any study drug in 35 (11%) patients in the avelumab maintenance group, 53 (16%) patients in the avelumab combination group, and 21 (6%) patients in the control group; the most common reasons (in at least three patients in any group) were diarrhoea (four [1%] patients),

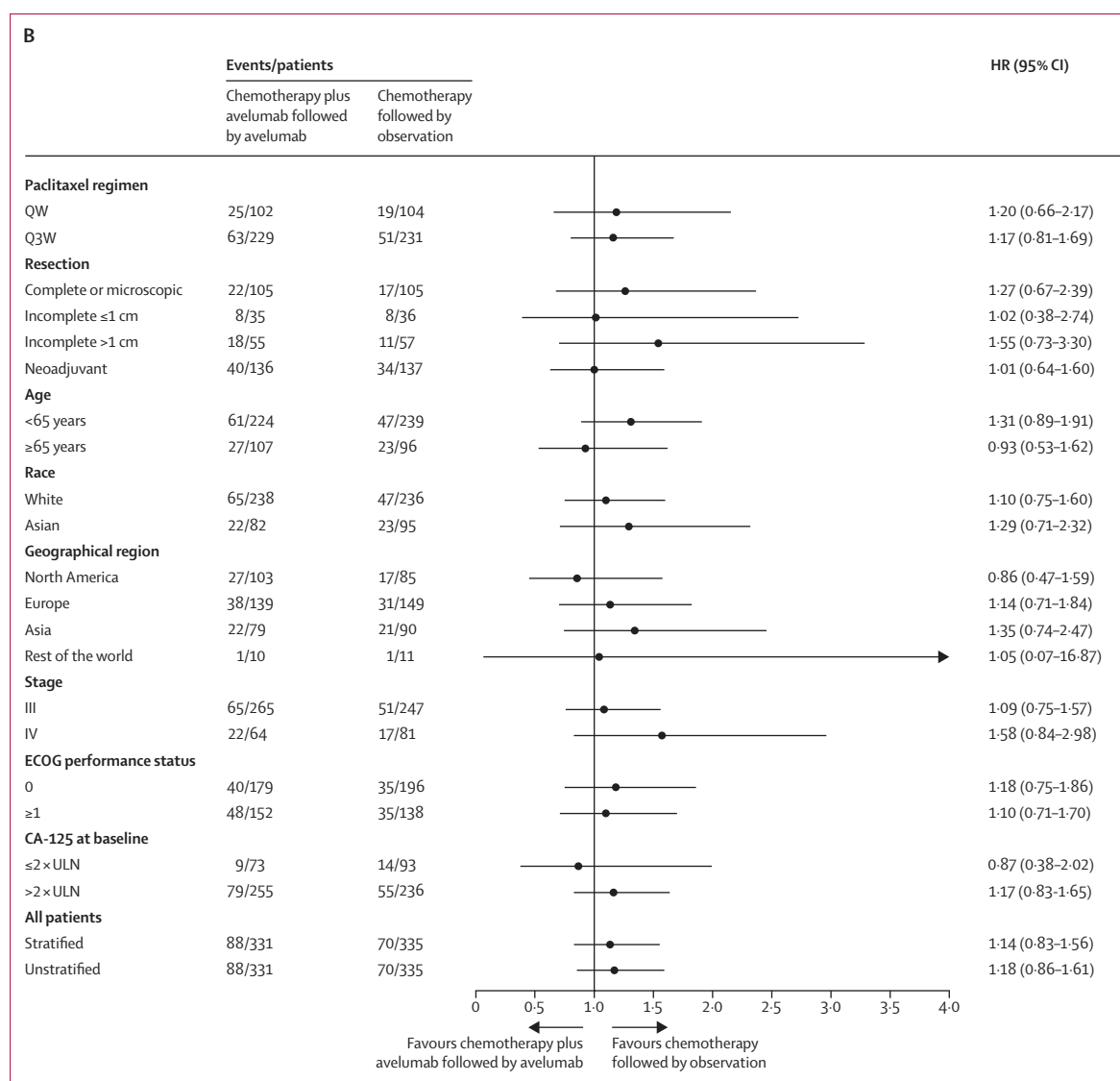


Figure 3: Forest plots of progression-free survival in baseline subgroups

Chemotherapy followed by avelumab (A) and chemotherapy plus avelumab followed by avelumab (B), each versus chemotherapy followed by observation. Except for the primary analysis (all patients), which was stratified according to randomisation stratification factors, all other analyses presented were unstratified. Any subgroups with less than 5% of patients are not included in the forest plot. Data for subgroups defined by ethnicity are not reported because more than 95% of the patient population were non-Hispanic or non-Latino. Data for subgroups defined by *BRCA* mutation status are not reported because the negative subgroup was more than 95% of the patient population. The data cutoff was Sept 7, 2018 (PD-L1 status data not shown due to later cutoff). ECOG=Eastern Cooperative Oncology Group. HR=hazard ratio. QW=once a week. Q3W=once every 3 weeks. ULN=upper limit of normal.

peripheral neuropathy (three [1%]), and anaemia (three [1%]) in the avelumab maintenance group; infusion-related reaction (single preferred term; five [2%]), alanine aminotransferase increase (four [1%]), thrombocytopenia (four [1%]), neutrophil count decrease (three [1%]), platelet count decrease (three [1%]), peripheral neuropathy (three [1%]), and peripheral sensory neuropathy (three [1%]) in the avelumab combination group; and peripheral sensory neuropathy (five [1%]) and peripheral neuropathy (three [1%]) in the control group.

Treatment-related deaths occurred in one (<1%) patient in the avelumab maintenance group (due to atrial

fibrillation) and one (<1%) patient in the avelumab combination group (due to disease progression). In the safety analysis, the total number of deaths in treated patients irrespective of relationship to study treatment was 34 (10%) in the avelumab maintenance group, 31 (9%) in the avelumab combination group, and 20 (6%) in the control group. Considering causes of death (more than one cause possible per patient), reasons included disease progression (29 [9%], 26 [8%], and 17 [5%], respectively), adverse events not related to study treatment (five [2%], eight [2%], and two [1%]), study treatment toxicity (one [<1%], 0, and 0), other reasons

(two [1%], three [1%], and one [$<1\%$]), and unknown reasons (one [$<1\%$], two [1%], and three [1%]).

As of Sept 7, 2018, in the avelumab maintenance, avelumab combination, and control groups, immune-related adverse events of any grade occurred in 53 (16%) of 328 patients, 92 (28%) of 329, and 0 of 334, respectively (appendix p 22). In the avelumab maintenance and

avelumab combination groups, grade 3–5 immune-related adverse events occurred in ten (3%) and 24 (7%) patients, and led to discontinuation of any study drug in eight (2%) and 19 (6%) patients. No deaths were attributed to immune-related adverse events. Infusion-related reactions of any grade occurred in 58 (18%) patients in the avelumab maintenance group, 65 (20%)

	Chemotherapy followed by avelumab (n=328)				Chemotherapy plus avelumab followed by avelumab (n=329)				Chemotherapy followed by observation (n=334)			
	Grade 1–2	Grade 3	Grade 4	Grade 5	Grade 1–2	Grade 3	Grade 4	Grade 5	Grade 1–2	Grade 3	Grade 4	Grade 5
Patients with treatment-emergent adverse event	100 (30%)	151 (46%)	67 (20%)	5 (2%)	90 (27%)	148 (45%)	84 (26%)	6 (2%)	111 (33%)	131 (39%)	76 (23%)	3 (1%)
Alopecia	165 (50%)	0	2 (1%)	0	169 (51%)	0	0	0	174 (52%)	2 (1%)	1 ($<1\%$)	0
Nausea	147 (45%)	6 (2%)	0	0	145 (44%)	7 (2%)	0	0	147 (44%)	5 (1%)	0	0
Fatigue	117 (36%)	6 (2%)	0	0	102 (31%)	12 (4%)	1 ($<1\%$)	0	98 (29%)	12 (4%)	0	0
Constipation	111 (34%)	3 (1%)	0	0	101 (31%)	0	1 ($<1\%$)	0	93 (28%)	3 (1%)	0	0
Peripheral sensory neuropathy	91 (28%)	0	0	0	76 (23%)	0	0	0	82 (25%)	0	0	0
Anaemia	82 (25%)	69 (21%)	0	0	92 (28%)	61 (19%)	2 (1%)	0	90 (27%)	53 (16%)	0	0
Vomiting	78 (24%)	9 (3%)	0	0	72 (22%)	8 (2%)	0	0	61 (18%)	7 (2%)	0	0
Diarrhoea	78 (24%)	8 (2%)	0	0	97 (29%)	6 (2%)	0	0	57 (17%)	7 (2%)	0	0
Arthralgia	75 (23%)	1 ($<1\%$)	0	0	81 (25%)	5 (2%)	0	0	56 (17%)	1 ($<1\%$)	0	0
Myalgia	66 (20%)	1 ($<1\%$)	0	0	51 (16%)	2 (1%)	0	0	40 (12%)	3 (1%)	0	0
Abdominal pain	65 (20%)	5 (2%)	0	0	64 (19%)	6 (2%)	0	0	52 (16%)	8 (2%)	0	0
Decreased appetite	63 (19%)	1 ($<1\%$)	0	0	54 (16%)	1 ($<1\%$)	0	0	36 (11%)	1 ($<1\%$)	0	0
Neuropathy peripheral	62 (19%)	1 ($<1\%$)	0	0	74 (22%)	3 (1%)	0	0	64 (19%)	1 ($<1\%$)	0	0
Rash	57 (17%)	2 (1%)	0	0	60 (18%)	6 (2%)	0	0	24 (7%)	0	1 ($<1\%$)	0
Headache	55 (17%)	1 ($<1\%$)	0	0	50 (15%)	1 ($<1\%$)	0	0	30 (9%)	0	0	0
Insomnia	50 (15%)	2 (1%)	0	0	39 (12%)	0	0	0	30 (9%)	2 (1%)	0	0
Dizziness	44 (13%)	0	1 ($<1\%$)	0	37 (11%)	1 ($<1\%$)	0	0	28 (8%)	0	0	0
Pruritus	37 (11%)	1 ($<1\%$)	0	0	36 (11%)	1 ($<1\%$)	0	0	19 (6%)	0	0	0
Pyrexia	37 (11%)	0	0	0	48 (15%)	1 ($<1\%$)	0	0	23 (7%)	1 ($<1\%$)	0	0
Dyspnoea	36 (11%)	3 (1%)	1 ($<1\%$)	0	45 (14%)	4 (1%)	0	0	29 (9%)	1 ($<1\%$)	0	0
Cough	36 (11%)	1 ($<1\%$)	0	0	55 (17%)	0	0	0	22 (7%)	0	0	0
Thrombocytopenia	33 (10%)	13 (4%)	1 ($<1\%$)	0	37 (11%)	16 (5%)	10 (3%)	0	41 (12%)	15 (4%)	3 (1%)	0
Urinary tract infection	33 (10%)	4 (1%)	0	0	42 (13%)	8 (2%)	0	0	27 (8%)	2 (1%)	0	0
Hypothyroidism	33 (10%)	1 ($<1\%$)	0	0	33 (10%)	0	0	0	5 (1%)	0	0	0
Hypomagnesaemia	31 (9%)	1 ($<1\%$)	0	0	38 (12%)	3 (1%)	1 ($<1\%$)	0	27 (8%)	0	0	0
Abdominal pain upper	31 (9%)	1 ($<1\%$)	0	0	38 (12%)	2 (1%)	0	0	24 (7%)	0	0	0
Asthenia	30 (9%)	5 (2%)	0	0	44 (13%)	2 (1%)	0	0	21 (6%)	1 ($<1\%$)	0	0
Back pain	29 (9%)	2 (1%)	0	0	34 (10%)	2 (1%)	0	0	28 (8%)	3 (1%)	0	0
Pain in extremity	29 (9%)	1 ($<1\%$)	0	0	32 (10%)	0	0	0	37 (11%)	0	0	0
Alanine aminotransferase increased	27 (8%)	1 ($<1\%$)	0	0	28 (9%)	5 (2%)	0	0	17 (5%)	3 (1%)	0	0
Neutropenia	23 (7%)	54 (16%)	37 (11%)	0	26 (8%)	54 (16%)	45 (14%)	0	25 (7%)	48 (14%)	40 (12%)	0
Platelet count decreased	18 (5%)	6 (2%)	1 ($<1\%$)	0	28 (9%)	8 (2%)	3 (1%)	0	29 (9%)	14 (4%)	1 ($<1\%$)	0
Hypokalaemia	16 (5%)	7 (2%)	0	0	15 (5%)	9 (3%)	3 (1%)	0	15 (4%)	4 (1%)	1 ($<1\%$)	0
Leukopenia	15 (5%)	12 (4%)	1 ($<1\%$)	0	17 (5%)	10 (3%)	1 ($<1\%$)	0	15 (4%)	5 (1%)	0	0
Hypertension	12 (4%)	4 (1%)	0	0	12 (4%)	7 (2%)	0	0	10 (3%)	3 (1%)	0	0
Rash maculopapular	12 (4%)	1 ($<1\%$)	0	0	16 (5%)	5 (2%)	0	0	5 (1%)	0	0	0
Hypertriglyceridaemia	12 (4%)	0	0	0	10 (3%)	5 (2%)	0	0	8 (2%)	1 ($<1\%$)	0	0
Neutrophil count decreased	11 (3%)	31 (9%)	18 (5%)	0	9 (3%)	29 (9%)	16 (5%)	0	16 (5%)	32 (10%)	27 (8%)	0

(Table 2 continues on next page)

	Chemotherapy followed by avelumab (n=328)				Chemotherapy plus avelumab followed by avelumab (n=329)				Chemotherapy followed by observation (n=334)			
	Grade 1–2	Grade 3	Grade 4	Grade 5	Grade 1–2	Grade 3	Grade 4	Grade 5	Grade 1–2	Grade 3	Grade 4	Grade 5
(Continued from previous page)												
Hyperglycaemia	10 (3%)	4 (1%)	1 (<1%)	0	2 (1%)	3 (1%)	1 (<1%)	0	4 (1%)	2 (1%)	1 (<1%)	0
White blood cell count decreased	9 (3%)	22 (7%)	1 (<1%)	0	13 (4%)	15 (5%)	3 (1%)	0	17 (5%)	15 (4%)	2 (1%)	0
Blood creatine phosphokinase increased	7 (2%)	2 (1%)	1 (<1%)	0	4 (1%)	4 (1%)	1 (<1%)	0	2 (1%)	1 (<1%)	0	0
Ascites	6 (2%)	6 (2%)	0	0	4 (1%)	4 (1%)	0	0	4 (1%)	2 (1%)	0	0
Lymphocyte count decreased	4 (1%)	4 (1%)	1 (<1%)	0	4 (1%)	6 (2%)	0	0	2 (1%)	3 (1%)	1 (<1%)	0
Hyponatraemia	4 (1%)	4 (1%)	0	0	3 (1%)	5 (2%)	0	0	0	3 (1%)	0	0
γ-glutamyltransferase increased	4 (1%)	3 (1%)	0	0	9 (3%)	5 (2%)	2 (1%)	0	9 (3%)	4 (1%)	0	0
Intestinal obstruction	3 (1%)	1 (<1%)	2 (1%)	0	1 (<1%)	6 (2%)	1 (<1%)	0	0	7 (2%)	0	0
Small intestinal obstruction	3 (1%)	1 (<1%)	0	0	1 (<1%)	7 (2%)	0	0	1 (<1%)	0	0	0
Febrile neutropenia	2 (1%)	11 (3%)	0	0	0	11 (3%)	2 (1%)	0	1 (<1%)	9 (3%)	1 (<1%)	0
Haemoglobin decreased	2 (1%)	4 (1%)	1 (<1%)	0	1 (<1%)	2 (1%)	0	0	1 (<1%)	2 (1%)	0	0
Lipase increased	1 (<1%)	4 (1%)	2 (1%)	0	4 (1%)	6 (2%)	1 (<1%)	0	1 (<1%)	2 (1%)	0	0
Pulmonary embolism	0	7 (2%)	2 (1%)	2 (1%)	2 (1%)	5 (2%)	1 (<1%)	0	3 (1%)	3 (1%)	1 (<1%)	1 (<1%)

Data are reported by number of patients: n (%). Grade 1–2 adverse events occurring in at least 10% of patients and grade 3–5 adverse events occurring in at least 2% of patients in any group are shown. A table showing adverse events of grade 1–2 occurring in at least 10% of patients and all adverse events of grade 3–5 is included in the appendix (pp 14–21).

Table 2: Treatment-emergent adverse events

patients in the avelumab combination group, and 44 (13%) patients in the control group (appendix p 22). Grade 3–5 infusion-related reactions occurred in two (1%), six (2%), and six (2%) patients, respectively. Discontinuation of any study drug due to infusion-related reactions occurred in three (1%), seven (2%), and four (1%) patients, respectively.

Discussion

The JAVELIN Ovarian 100 trial did not meet either of its two primary objectives of improving progression-free survival with two avelumab regimens with chemotherapy versus chemotherapy alone. At the interim analysis, test statistics in both avelumab groups had crossed the prespecified futility boundaries and the trial was stopped. Stratified HRs for progression-free survival favoured the control group, indicating an observed detrimental effect in both avelumab groups. Overall survival data were immature. No benefit was observed in either experimental group compared with the control group in terms of objective response rate, maintenance progression-free survival, or pathological complete response. The safety profiles for avelumab in the experimental groups were broadly similar to the profile for chemotherapy alone, with an increase in frequency of serious adverse events, slight increases in some types of adverse event, and low occurrence of immune-related adverse events, consistent with the known safety profile of avelumab monotherapy.²⁷ No new safety signals were identified for avelumab

administered in combination with and following chemotherapy or as maintenance therapy alone.

Exploratory subgroup analyses based on baseline characteristics and stratification factors did not identify subsets of patients with clear progression-free survival benefit in either avelumab group. Additionally, PD-L1 status did not predict benefit with avelumab treatment, either as maintenance therapy or in combination with chemotherapy, which is in contrast with findings from the phase 3 JAVELIN Ovarian 200 trial of avelumab as monotherapy or in combination with PLD versus PLD alone in patients with platinum-resistant or platinum-refractory epithelial ovarian cancer.²² Although the JAVELIN Ovarian 200 trial did not meet its primary objectives of significantly improving progression-free survival or overall survival in the overall population, biomarker analyses indicated that PD-L1 status might predict benefit with avelumab plus PLD versus PLD alone. The absence of a potential predictive role of PD-L1 status in the current trial might be due to the differences in tumour biology or microenvironment or different immunological effects of chemotherapies^{28,29} administered in patients with previously untreated epithelial ovarian cancer versus platinum-resistant or platinum-refractory epithelial ovarian cancer; as a result, tumours that have recurred after frontline chemotherapy and are PD-L1-positive might be more sensitive to subsequent combination treatment with chemotherapy and immune checkpoint inhibitors than PD-L1-positive treatment-naïve tumours.

In 2020, a phase 3 randomised trial of a different anti-PD-L1 antibody, atezolizumab, administered with bevacizumab, paclitaxel, and carboplatin in patients with newly diagnosed advanced epithelial ovarian cancer (IMagyn050), also did not meet one of its primary objectives of improved progression-free survival with the atezolizumab combination regimen versus placebo, bevacizumab, paclitaxel, and carboplatin.³⁰ Data for the other primary endpoint of overall survival are immature, and follow-up is ongoing. Results from our trial and IMagyn050 suggest that the addition of an immune checkpoint inhibitor to frontline chemotherapy does not improve efficacy in an overall population. The negative outcome of our trial was unexpected and there is no obvious explanation for the results. Several other phase 3 studies investigating the activity of immune checkpoint inhibitors in combination with chemotherapy, bevacizumab, or poly-ADP ribose polymerase (PARP) inhibitors in the frontline setting for advanced epithelial ovarian cancer are in progress. These studies include the DUO-O trial of durvalumab (anti-PD-L1) plus chemotherapy and bevacizumab followed by durvalumab plus bevacizumab and olaparib (PARP inhibitor) maintenance (NCT03737643); the ENGOT-OV44 trial of dostarlimab (anti-PD-1) plus chemotherapy and niraparib (PARP inhibitor; NCT03602859); and the ENGOT-OV43 trial of pembrolizumab plus chemotherapy followed by olaparib (NCT03740165). These ongoing trials will hopefully provide further clarity on whether immune checkpoint inhibitors have any role in the frontline treatment of patients with epithelial ovarian cancer.

This trial had several limitations. Firstly, no predictive biomarkers were available to aid patient selection for the trial. Secondly, baseline data on *BRCA* status were not systematically collected during the trial; therefore, the association between *BRCA* status and outcomes could not be evaluated. Similarly, patients were not assessed for homologous recombination deficiency, which has recently become a biomarker of interest in the treatment of patients with epithelial ovarian cancer via other drugs. Additionally, data on second-line therapies were not collected in most patients because the trial was terminated at the interim analysis. Finally, longer term efficacy data were not obtained because, when the trial was stopped after the interim analysis, maintenance treatment was discontinued and long-term follow-up was not done, consistent with the recommendations of the independent data monitoring committee and protocol steering committee.

In conclusion, the JAVELIN Ovarian 100 trial showed that avelumab as a maintenance therapy or in combination with chemotherapy did not improve progression-free survival compared with chemotherapy alone in patients with previously untreated epithelial ovarian cancer.

Contributors

BJM, MJB, RAS, and JAL contributed to study design. BJM, NC, AMO, KF, MJB, LR, EVP, GS, YVS, MCL, SMB, JS, KY, RAS, CL, and JAL

contributed to data collection. BJM, RAS, XZ, and CL contributed to data analysis. BJM, AMO, LR, RAS, XZ, JPS, CL, and JAL contributed to data interpretation. All authors had access to the data reported, and the lead and senior authors (BJM and JAL) and co-authors who were employees of the sponsor (RAS, XZ, JPS, and CL) had access to the raw data. BJM, XZ, and JAL accessed and verified the data. All authors contributed to manuscript writing and subsequent drafts and provided final approval to submit the manuscript for publication.

Declaration of interests

BJM reports receiving honoraria from and serving as a consultant or advisor for Agenus, Akeso, Aravive, AstraZeneca, Clovis Oncology, Eisai, Elevar Therapeutics, Genmab and Seattle Genetics, GOG Foundation, Gradalis, ImmunoGen, Karyopharm Therapeutics, Iovance Biotherapeutics, Merck & Co, US Oncology Network (owned by McKesson), Mersana Therapeutics, Novocure, Myriad Genetics, Pfizer, Puma Biotechnology, Genentech (subsidiary of Roche), Sorrento Therapeutics, Tesaro (owned by GlaxoSmithKline), and VBL Therapeutics. NC reports receiving honoraria from and serving as a consultant or advisor for Amgen, AstraZeneca, Clovis Oncology, GlaxoSmithKline, Pfizer, Roche, and Tesaro; receiving honoraria from Novartis; serving as a consultant or advisor for Advaxis, BioCad, ImmunoGen, OncXerna, Merck Sharp & Dohme, and Takeda; and participating on a data safety monitoring board or advisory board for Advaxis, Amgen, AstraZeneca, BioCad, Clovis Oncology, GlaxoSmithKline, ImmunoGen, Merck Sharp & Dohme, OncXerna, Pfizer, Roche, Takeda, and Tesaro. KF reports receiving research funding and reimbursement for travel, accommodation, and expenses from Pfizer. MJB reports participating on a data safety monitoring board or advisory board for AstraZeneca, Clovis Oncology, GlaxoSmithKline, OncoQuest, and VBL Therapeutics. LR reports receiving honoraria from AstraZeneca, GlaxoSmithKline, and Merck & Co; serving as a consultant or advisor for Agenus, Akeso Bio, AstraZeneca, Clovis Oncology, EMD Serono, GlaxoSmithKline, Genentech, Intuitive Surgical, Myriad, Novartis, and Rubius; and having received institutional research grants from Akeso Biopharma, AstraZeneca, Genentech, GOG Foundation, GlaxoSmithKline, Incyte, Karyopharm, Merck & Co, Mersana Therapeutics, On Target Laboratories, and Seagen. GS reports receiving honoraria from Clovis Oncology; serving as a consultant or advisor for Johnson & Johnson and Tesaro; and having received research funding from Merck Sharp & Dohme. JS has received institutional research funding from AstraZeneca, Boehringer Ingelheim, Daiichi Sankyo, GlaxoSmithKline, Lilly, Merck Sharp & Dohme, Novartis, Pfizer, Roche, and Sanofi. KY reports receiving honoraria from AstraZeneca, Chugai Pharma, Eisai, Novartis, Pfizer, and Takeda; and serving as a consultant or advisor for AstraZeneca, Chugai Pharma, Eisai, Novartis, Ono Pharmaceutical, and Takeda. RAS reports employment at Pfizer at the time the study was done and owns stock in Pfizer. XZ reports employment at and owns stock in Pfizer. JPS reports employment at and owns stock in Pfizer. CL reports employment at Pfizer and owns stock in Eli Lilly and Pfizer. JAL reports receiving honoraria from AstraZeneca, GlaxoSmithKline, and Pfizer; and is the vice president of the European Society of Gynaecological Oncology and an editor of the gynaecological clinical practice guidelines for the European Society for Medical Oncology. AMO, EVP, YVS, MCL, and SMB declare no competing interests.

Data sharing

On request, and subject to certain criteria, conditions, and exceptions (see online), Pfizer will provide access to individual de-identified participant data from Pfizer-sponsored global interventional clinical studies done for medicines, vaccines, and medical devices: for indications that have been approved in the USA or EU, or both; or in programmes that have been terminated (ie, development for all indications has been discontinued). Pfizer will also consider requests for the protocol, data dictionary, and statistical analysis plan. Data can be requested from Pfizer trials 24 months after study completion. The de-identified participant data will be made available to researchers whose proposals meet the research criteria and other conditions, and for which an exception does not apply, via a secure portal. To gain access, data requestors must enter into a data access agreement with Pfizer.

Acknowledgments

This study was sponsored by Pfizer as part of an alliance between Pfizer and Merck KGaA, Darmstadt, Germany. Medical writing support was

provided by Amy Davidson of ClinicalThinking and was funded by Pfizer and Merck KGaA. The authors would like to thank the patients and their families, investigators, coinvestigators, and the study teams at each of the participating centres. At the time the study was done, Arizona Oncology Associates PC-HAL was the affiliation of SMB and Pfizer Oncology was the affiliation of RAS.

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